

What is NTP ?

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NTP (Network Time Protocol) is a protocol used to synchronize computer clocks over a network.

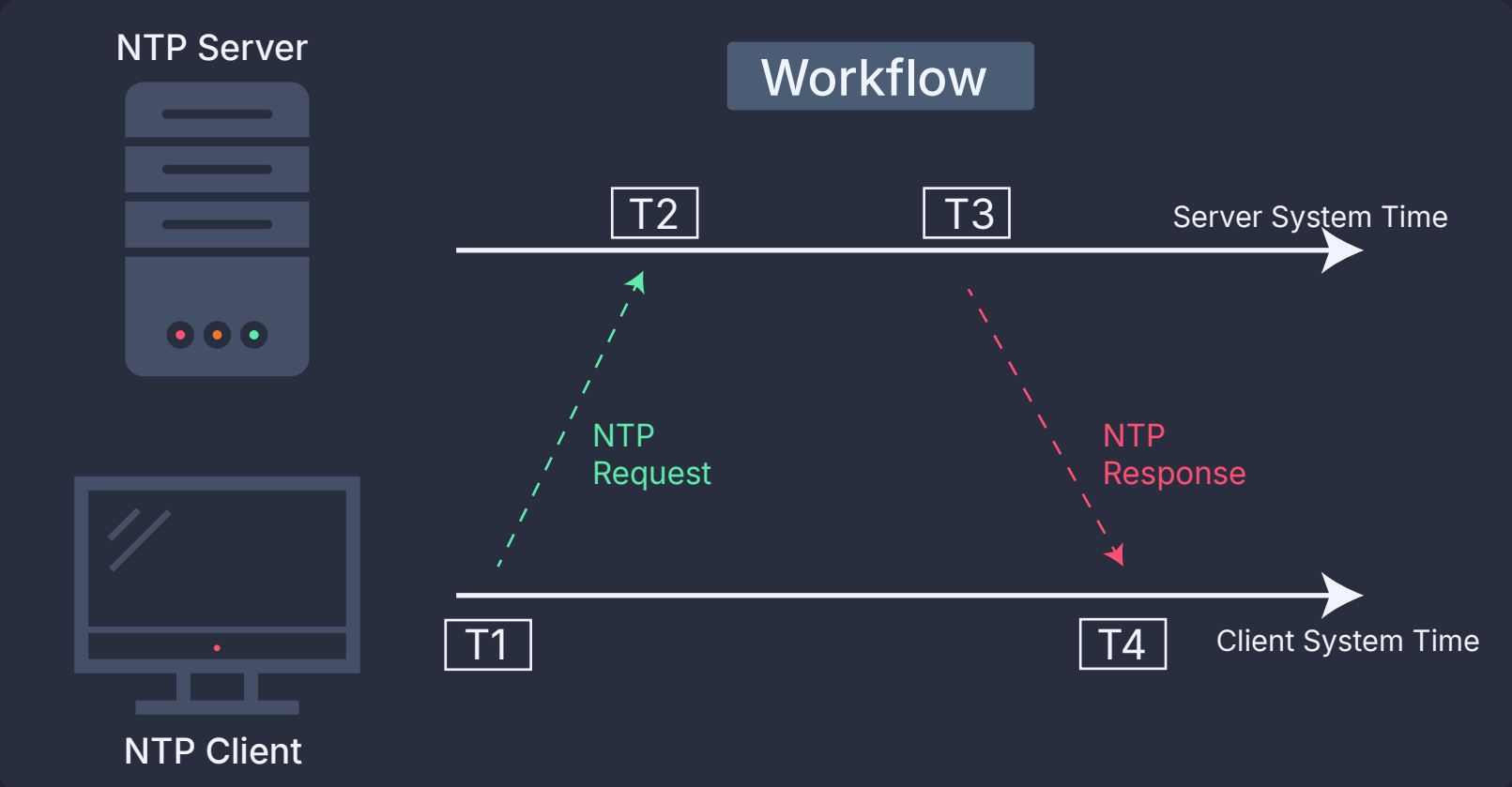
It accounts for network delays and adjusts your local system clock to match an accurate reference clock (usually from an NTP server).

UDP Port: NTP uses UDP on port 123.

Key Terms & Timeline

Timestamps

- T1:** The time (according to the client) when the request is sent.
- T2:** The time (according to the server) when the request is received.
- T3:** The time (according to the server) when the response is sent.
- T4:** The time (according to the client) when the response is received.



Network Delay (D)

$$D = (T4 - T1) - (T3 - T2)$$

- (T4 - T1) is the total round-trip time (client perspective).
- (T3 - T2) is the server's processing time.
- Subtracting the server processing time from the total round trip gives an estimate of network latency alone.

Clock Offset (θ)

$$\theta = \frac{(T2 - T1) + (T3 - T4)}{2}$$

- If θ is positive, your clock is behind the server and needs to move forward.
- If θ is negative, your clock is ahead of the server and needs to slow down or move backward.

Modes of Operation:

Client/Server

Multicast

Symmetric(Peer)

Broadcast



Clock Error Correction

Large Offset (≥ ~128ms):

- The client performs a step adjustment: it jumps directly to the correct time (current time + offset).
- This is a fast correction but can cause abrupt changes.

Small Offset (< ~128ms):

- The client gradually speeds up or slows down its local clock.
- This smooth correction avoids sudden time jumps.

Network Quality Check

If the network delay (D) is extremely high or unreliable:

- The NTP client may ignore or reduce the influence of that server's time.
- This prevents large, inaccurate shifts in clock synchronization.

NTP Server Hierarchy

